USN											

24CS340

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester: CIE-2

COMPUTER ORGANIZATION

B, C & D Sections

Duration (In hrs): 1.15 Hrs Date: 09/01/2026 Max. Marks: 30

NOTE: Question 1 is compulsory, Question 2 and 3 has internal choice.

Q.No	CO	CD	PI	QUESTION	MARKS
1.	CO4	L3	1.4.1	a.Draw 4- bit carry look ahead adder and explain its operation with expressions. b.Perform the binary multiplication of (+14) and (-7) using Booth algorithm and 3-bit pair recoding.	9+6=15
2.	CO3	L2	2.3.1	a. Explain the structure of Static RAM cell with diagram. b. Briefly explain Direct Mapping Technique. c.Write the internal organization of 256M bit DRAM chip(32Mx8)	4+4+7 = 15
				OR	
3.	CO3	L2	2.3.1	a. Cache : 128 blocks : 16 words per blocks : 2048 words Main memory : 4k blocks : 16 words per block : 64k words i)Calculate the bits in MM address ii)Calculate the size of various fields in a)Direct b)2-way set c)4-way set d) Fully associative b. Explain the following terminologies i) Cache Hit ii) Cache miss iii) Replacement Algorithm	10+5 = 15

CO	Course Outcomes	Cognitive Domain
CO2	Explain the different ways of communicating with the I/O devices.	L2
CO3	Design and analyse the performance of memory systems.	L3

PI	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.3.1	Able to apply computer engineering principles to formulate modules of a system with required applicability and performance